

**TABLE III**  
**Vision Z Microscope**

| Specifications      |                                              |
|---------------------|----------------------------------------------|
| Resolution          | HD 720p (1280×720)                           |
| PC connections      | HDMI out                                     |
| Focus               | Auto/manual                                  |
| Illumination        | Adjustable LED illumination                  |
| Dimensions          | Bench-top stand:<br>410mm x 240mm x<br>350mm |
| Weight              | Bench-top stand: 1.1kg                       |
| Power supply        | 100/240V, 50/60Hz                            |
| Power consumption   | 8 watts                                      |
| Recommended monitor | 16:9, DVI & HDMI                             |
| Response time       | 5ms, Signal: HD 720p                         |

**TABLE IV**  
**Objective Lenses**

| Lens | Magnification Range | Working Distance | Field of View (Maximum) |
|------|---------------------|------------------|-------------------------|
| 3D   | 1.25x to 35x        | 333mm            | 294mm                   |
| 4D   | 2.0x to 55x         | 250mm            | 219mm                   |
| 5D   | 2.5x to 60x         | 200mm            | 174mm                   |
| 10D  | 5.5x to 120x        | 100mm            | 87mm                    |

ing with zoom magnification options up to 120x, making VisionZ ideal for a very wide range of precision magnification tasks.

**Manufacturer** – Olympus

**Model** – DSX500

The DSX500 is a high-resolution upright motorised microscope with 13x zoom optics.



Fig. 2: DSX500 microscope

**TABLE V**  
**DSX500 Microscope**

| Specification                  |                                                                                                                                                                   |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zoom Ratio                     | 13.5x optical zoom, 30x with digital zoom                                                                                                                         |
| Accuracy of Magnification      | ±3%                                                                                                                                                               |
| Repeatability of Magnification | 3σn-1 = ±2%                                                                                                                                                       |
| Illumination                   | Bright field: LED, Dark field: LED                                                                                                                                |
| Optional Illumination          | High-intensity LED                                                                                                                                                |
| Image Sensor                   | Colour CCD (total pixels: 2.10 megapixels)<br>Total pixels: 1688 (H) x 1248 (V)<br>Available pixels: 1628 (H) x 1236 (V)<br>Effective pixels: 1600 (H) x 1200 (V) |
| Cooling Method                 | Peltier cooling                                                                                                                                                   |
| Scan Mode                      | Progressive scan                                                                                                                                                  |
| Frame Rate                     | 15fps/27fps with binning mode                                                                                                                                     |
| Image Resolution               | Normal: 1194 × 1194(1:1)/1592 × 1194(4:3)<br>Fine: 1194 × 1194(1:1)/1592 × 1194(4:3)<br>Superfine: 3594 × 3594(1:1)/4792 × 3594(4:3)                              |
| Sensitivity                    | ISO100/200/400/800/1600 equivalent                                                                                                                                |
| LCD Monitor Size               | 584.2mm (23-inch) with touch panel and full-HD colour LCD monitor                                                                                                 |
| LCD Monitor Resolution         | 1920 (H) x 1080 (V)                                                                                                                                               |
| Weight                         | Approximately 38.6kg (Main frame, Motorised stage, LCD Monitor, Control box, Controller)                                                                          |
| Input Rating                   | 100 to 120V/220 to 240V, 185VA, 50/60Hz                                                                                                                           |

**TABLE VI**  
**Objective Lenses**

| Lens         | Magnification Range | Working Distance | Actual Field of View |
|--------------|---------------------|------------------|----------------------|
| MPLFLN1.25X  | 17x to 225x         | 3.5mm            | 15,684 to 1206 μm    |
| MPLFLN2.5X   | 35x to 451x         | 10.7mm           | 7842 to 603μm        |
| MPLFLN5XBDP  | 69x to 901x         | 12mm             | 3921 to 302μm        |
| MPLFLN10XBDP | 139x to 1803x       | 6.5mm            | 1960 to 151μm        |
| MPLFLN20XBDP | 277x to 3606x       | 3mm              | 980 to 75μm          |
| MPLFLN50XBDP | 693x to 9014x       | 1mm              | 392 to 30μm          |
| LMPLFLN10XBD | 139x to 1803x       | 10mm             | 1960 to 151μm        |
| LMPLFLN20XBD | 277x to 3606x       | 12mm             | 980 to 75μm          |
| LMPLFLN50XBD | 693x to 9014x       | 10.6mm           | 392 to 30μm          |
| MPLAPON50X   | 693x to 9014x       | 0.35mm           | 392 to 30μm          |

**Manufacturer** – Dino Lite

**Model** – AM4815T Dino-Lite Edge

Adding to the features of Dino-Lite Edge, the AM4815T provides image processing features, EDOF (extended depth of field) and EDR (extended dynamic range), for further extending its limitation over

depth of focus or dynamic range.



Fig. 3: AM4815T Dino-Lite Edge microscope